

ELESSANDRO PRESTES MACEDO

Software Engineer | PHP & Laravel | Backend & Full Stack | Architecture & Applied AI

Toledo, PR – Brazil

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PROFESSIONAL SUMMARY

Software Engineer with 9+ years building and modernizing mission-critical systems that can't go down or lose data (CAPES, ONS, Energy and Industry): 2,500+ RPS with <300ms latency serving 448+ institutions and 99.9% uptime. PHP (Laravel) and Node.js in Microservices and Event-Driven architectures on AWS and Azure. Legacy systems modernized with the Strangler Fig Pattern (56,400+ lines eliminated, 40% less rework) and applied AI (SDD, RAG, MCP).

TECHNICAL SKILLS

PHP | Laravel | MySQL/MariaDB | Redis | REST APIs | Git | SOLID | Legacy refactoring | Spec-Driven Development

Languages: PHP (Laravel, Symfony), Node.js (Express, NestJS), TypeScript, JavaScript

Databases: MySQL/MariaDB, PostgreSQL, Oracle (PL/SQL), Redis, MongoDB, SQL Server

Architecture: Microservices, Monolithic, Serverless, BFF, Event-Driven Architecture, Strangler Fig Pattern, TDD, SOLID, Clean Architecture

Messaging: RabbitMQ, Kafka, Amazon MQ

Testing: Pest, PHPUnit, Vitest

Frontend: Angular, Vue.js, React

Cloud & DevOps: Git, AWS (Lambda, ECS, SQS, CloudWatch), Azure (Monitor, Functions, IoT), Docker, Kubernetes, CI/CD (GitHub Actions, GitLab)

AI-Assisted Software Engineering: Claude Code, GitHub Copilot, Codex, Spec-Driven Development (SDD), RAG, MCP (Model Context Protocol)

IoT: Internet of Things (Industrial IoT)

PROFESSIONAL EXPERIENCE

EPM DEVTECH — Toledo, Paraná, Brazil

Technical Lead in Software Engineering – Full Stack & Systems Architecture (Jun 2026 – Present)

- **Architecture & Engineering:** Planning and development of modern websites, platforms and complete web systems, as well as scalable APIs with PHP/Laravel, Node.js, Express.js and NestJS, including integrations between enterprise systems (ERPs, CRMs and payment platforms).
- **End-to-End Full Stack Solutions:** Design and development of websites and web applications, working on the frontend with Vue.js, Angular or React and on the backend with PHP/Laravel and Node.js (Express.js and NestJS), plus database modeling, automated testing and continuous cloud deployment (CI/CD).
- **Modernization & Refactoring:** Redesign of corporate websites, updating of legacy systems, critical performance improvements and technical debt reduction through the Strangler Fig Pattern.
- **Technical Consulting:** Supporting companies in defining software architecture, engineering best practices and technical feasibility for new websites and systems.

Software Engineer with Applied AI (Oct 2025 – May 2026)

- **System Modernization and Migration:** Led the modernization of the legacy monolithic platform to PHP 8.2 and Laravel 12, applying the Strangler Fig Pattern to drive the architecture's evolution incrementally and without interruptions. Single-handedly drove the entire architecture, development and evolution of the application, eliminating more than 56,400 lines of legacy code and setting the foundation for scalable growth.
- **Automated Testing and Quality:** Built 2,399 automated tests across 241 files with Pest/PHPUnit, applying SOLID and Clean Code to raise software quality and reliability.
- **APIs and Integrations:** Evolved the platform to 384 routes/endpoints and 181 migrations, developing REST APIs, WhatsApp Business integrations and automations that reduced manual work by 35%.
- **Architecture and Rework Reduction:** Reduced rework by 40% through Software Architecture, automated testing, technical documentation and CI/CD with Docker and GitLab.
- **Applied AI and SDD:** Increased productivity and traceability by applying the Spec-Driven Development (SDD) methodology and AI-Assisted Software Engineering, using Claude Code, GitHub Copilot and Codex, along with RAG and MCP, for specification, implementation, testing, review and refactoring.

Datainfo (CAPES Project) — Brasília, Federal District, Brazil

Programmer Analyst / Tech Lead (Oct 2024 – Sep 2025)

- **Architectural Modernization (SIPREC):** Co-architected and executed the modernization of the SIPREC platform to absorb the national demand of programs such as UAB and AUXPE across 448+ higher education institutions, applying the Strangler Fig Pattern to incrementally migrate the legacy monolith to a microservices, BFF and distributed systems architecture, ensuring capacity for peaks of up to 10,000 concurrent users, 2,500 RPS throughput and average latency under 300ms.
- **SISCAD (CAPES) – Tech Lead:** Led and executed the ground-up development of SISCAD as Tech Lead and developer, automating the management of thousands of administrative collection cases and debt installment agreements through a resilient architecture built with PHP (Laravel), Angular, Oracle DB, Redis and asynchronous messaging with RabbitMQ.
- **DevOps and Reliability:** Increased operational reliability and reduced production failures by managing the infrastructure and deployment processes of critical environments with Docker, GitLab CI/CD and DevOps practices.
- **Quality Assurance:** Improved code quality indicators by 45% and reduced critical vulnerabilities by implementing continuous static analysis with SonarQube.
- **Productivity and Mentoring:** Increased feature delivery per sprint by 25%, leading the adoption of GitHub Copilot integrated with the SDD methodology across the team through technical mentoring and standardization of development-cycle specifications.

Energia Pecém — São Gonçalo do Amarante, Ceará, Brazil

Full Stack Developer (May 2023 – Jul 2024)

- **SIGMA Project – Operational Traceability:** Increased operational traceability by 40%, developing real-time equipment validation and tracking solutions with Node.js, Laravel, Vue.js and PostgreSQL.
- **Data Management:** Reduced operational failures by 35% and improved data consistency across the platform by designing real-time synchronization and state management flows with Redis.
- **Asynchronous Architecture:** Increased processing capacity by 50% and eliminated operational bottlenecks by designing an asynchronous, resilient Event-Driven Architecture with RabbitMQ and Laravel Jobs.
- **Cloud Monitoring:** Reduced critical production incidents by 45% and improved response time (MTTR) by implementing predictive monitoring and observability with Azure Monitor and Azure Communication Services.
- **Availability:** Ensured 99.9% platform availability (uptime), evolving the architecture with a focus on high scalability, decoupling and support for the operation's continuous growth.

AMcom (GENIN Project – ONS / Brazilian National Electric System Operator) — Blumenau, Santa Catarina, Brazil

Systems Developer (Jul 2022 – Apr 2023)

- **National Integrations:** Enabled the calculation of metrics for changing the national tariff flag in the Genin Project, ensuring 100% integrity and zero data loss in regulatory consolidation, through mission-critical integrations with Itaipu and INMET via PHP (Laravel), REST and SOAP.
- **Real-Time Processing:** Helped define the real-time architecture for energy and climate data at national scale, reducing data ingestion latency by 40% through decoupled microservices with PostgreSQL and Redis.
- **AWS Infrastructure:** Automated infrastructure and the continuous delivery pipeline on AWS, reducing deployment time by 60% and ensuring 99.9% service availability, with pipelines in CodeBuild, orchestration via API Gateway and proactive monitoring with CloudWatch.

Grupo Intellectus (Educational Project – Mato Grosso State Government) — Campo Grande, Mato Grosso do Sul, Brazil

Full Stack Developer (Oct 2021 – Jul 2022)

- **Serverless Architecture:** Helped design the architecture and developed the Online Chess platform on a Serverless Architecture on AWS, cutting costs by 35% and ensuring automatic scalability via AWS Lambda, SQS and SNS.
- **Web Development:** Worked on the development and integration of the platform for more than 650 state schools across 141 municipalities in Mato Grosso, connecting the solution to SEDUC-MT systems through Angular on the frontend and PHP (Laravel) APIs on the backend.
- **Observability:** Built the observability and centralized logging layer with ELK Stack, reducing the platform's mean time to diagnose incidents (MTTR) by 50%.
- **CI/CD:** Accelerated the delivery cycle of new features by 60%, automating continuous integration and delivery pipelines via GitHub Actions with automated code validations.
- **Availability:** Ensured 99.9% availability during peaks of simultaneous access at schools across the state, collaborating on System Design solutions focused on low latency, resilience and service decoupling.

Grupo Paraíso — Terra Roxa, Paraná, Brazil

Full Stack Developer (Aug 2016 – Sep 2021)

- **ERP Modernization:** Contributed to a 30% increase in operational efficiency across the finance, logistics and industrial areas, developing and evolving strategic modules of a corporate ERP with PHP (Laravel, Symfony), PostgreSQL and Oracle.
- **Frontend Development:** Developed frontend interfaces and modules with Angular, Vue.js and React, integrated with the APIs and services of the corporate ecosystem, serving different business operations, including ERP, e-commerce and industrial systems.

- **Legacy Modernization:** Helped reduce software maintenance costs by 40%, taking part in initiatives to migrate legacy monolithic systems to a microservices and distributed systems architecture with Node.js, using the Strangler Fig Pattern to enable gradual evolution without impacting operations.
- **Industrial IoT & Azure:** Helped reduce operational and human errors by 30%, taking part in the implementation of Industrial IoT integrations on the Azure Cloud ecosystem to connect production equipment to corporate systems.
- **Real-Time Communication:** Contributed to optimized, low-latency data synchronization by developing bi-directional real-time communication solutions based on WebSockets.
- **Event-Driven Architecture:** Helped ensure 99.9% availability by co-architecting scalable, event-driven solutions on AWS (Event-Driven Architecture) with Kafka, RabbitMQ, Redis and PostgreSQL.
- **Engineering Quality:** Worked with the team to raise the quality and maintainability of systems, applying patterns such as SOLID, Clean Architecture, TDD and event-driven architecture.

EDUCATION

Bachelor's Degree in Systems Analysis and Development – Universidade Paranaense (UNIPAR), Brazil (2012 – 2015).

LANGUAGES

Portuguese: Native.

English: Advanced in technical reading and writing; basic conversational level.